

How the Largest Economies Fund Debt Service: Interest, Refinancing, and Revenue in the Top Five Economies

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Abstract. The five largest economies by GDP pay for debt service almost entirely by refinancing maturing principal and covering interest from a mix of revenue and new borrowing. Using central-government fiscal series for the United States, China, Germany, and India, Japan Ministry of Finance budget data for Japan, and market data through September 2026, this paper measures how much of each government’s revenue goes to interest, how much of each year’s net new borrowing just pays the interest bill, and how much of gross issuance is refinancing. The 2021–2024 rate-hiking cycle doubled the US federal interest burden, from 10.9 percent of revenue in 2020 to 20.3 percent in 2024 (3.6 percent of GDP); 45 percent of 2023 US net borrowing just paid interest. India carries the heaviest load in the sample at 34.0 percent of revenue in 2022, and 93 percent of its 2015–2022 borrowing need was interest. Germany is the lightest at 2.5 percent of revenue. China is small at 3.4 percent of revenue in 2021, and its central bank eased while others tightened. Japan’s general-account interest bill, JPY 10.5 trillion in the FY2025 budget (9.2 percent of spending, about 1.7 percent of GDP), is projected by the Ministry of Finance to reach JPY 16.1 trillion by FY2028; roughly 83 percent of Japanese government bond issuance is refinancing. In the United States, 2024 gross federal issuance of USD 29.3 trillion against about USD 2 trillion of net new borrowing implies roughly 93 percent of gross issuance refinanced maturing principal. Data vintage: September 2026. World Bank series cover the central government only and have material gaps for China and Japan, documented in Section 6.

Keywords: sovereign debt, debt service, interest burden, fiscal space, refinancing

JEL codes: E62, H63, H68

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1 Introduction

How do the largest economies pay for their debt? The question has two parts. First, interest: what share of government revenue is consumed by interest payments, and what share of each year's new borrowing exists only to pay interest? Second, principal: when bonds mature, is the principal repaid from revenue or refinanced with new bonds? The answers determine fiscal space, which is the room a government has to fund everything else it does.

The question matters now because the global rate-hiking cycle of 2021–2024 repriced sovereign borrowing costs after a decade of near-zero rates. Interest bills that were flat for years are rising, and the rise lands unevenly: it depends on debt stocks, maturity structures, and how much of the budget was already committed to interest. This paper measures the funding of debt service for the five largest economies by nominal GDP, the United States, China, Germany, Japan, and India, over 2000–2026 where data permit.

This paper does three things. It assembles central-government fiscal series from the World Bank's World Development Indicators (WDI), retrieved in September 2026, and computes a consistent set of debt-service metrics: interest as a share of revenue, interest as a share of GDP, the primary balance, and the share of net new borrowing absorbed by interest. It adds market context from 10-year government bond yields and debt-scale benchmarks. And it treats Japan separately using Ministry of Finance (MOF) budget documents, because WDI carries no Japanese central-government interest series and Japan's debt dynamics are extreme enough to need their own accounting.

The preview of results: the US federal interest burden doubled between 2020 and 2024, from 10.9 to 20.3 percent of revenue, and 45 percent of 2023 net borrowing just paid interest. India has the heaviest burden in the sample: 34.0 percent of revenue in 2022, with 93 percent of its 2015–2022 borrowing need consisting of interest. Germany has the lightest: 2.5 percent of revenue. China's burden is small, 3.4 percent of revenue in 2021, and its central bank moved the opposite direction from Western central banks. Japan's interest bill is snapping back after three decades of decline: JPY 10.5 trillion in the FY2025 initial budget, JPY 13.0 trillion in the FY2026 budget, a MOF projection of JPY 16.1 trillion by FY2028, and a Fiscal System Council high-yield scenario of JPY 45.2 trillion by FY2035. Across all five, maturing principal is overwhelmingly refinanced rather than repaid: in the US, about 93 percent of 2024 gross issuance was refinancing.

Section 2 describes how sovereign funding works. Section 3 documents the data. Section 4 gives the exact computations. Section 5 presents results. Section 6 states the limitations. Section 7 concludes. A data appendix lists every series and source.

2 Institutional background

Sovereign debt service has two components: interest on outstanding debt, and principal that matures and must be repaid or refinanced. In practice, large sovereign borrowers almost never repay principal from revenue. They refinance it: when a bond matures, they issue a new bond and use the proceeds to pay the old one. The standard measure is therefore gross issuance versus net issuance. Gross issuance is everything sold in a year, including bonds that replace maturing ones. Net issuance is gross minus maturing principal, which equals the year's new borrowing. When gross issuance is many times net issuance, the government is rolling its debt stock, and the interest bill is the part of debt service that actually consumes fiscal resources.

Interest, unlike principal, must be paid from revenue or from additional borrowing. Two ratios capture the burden. Interest as a share of revenue measures how much of the budget is pre-committed before any policy choice is made. Interest as a share of GDP scales the burden to the economy. A third measure, the share of net new borrowing absorbed by interest, answers a narrower question: of the new debt issued in a year, how much exists only to pay interest? When that share approaches 100 percent, the government’s entire deficit is the interest bill, and any primary deficit, spending beyond interest in excess of revenue, adds on top.

The primary balance, revenue minus non-interest spending, separates the interest burden from current policy choices. A government can run a primary surplus and still borrow, if interest exceeds that surplus. India has done exactly this for most of the sample period.

One definitional point matters throughout. The WDI fiscal series used here cover the *central* government, not general government. For the United States that means the federal government, excluding states and municipalities. For Germany it means the Bund, excluding Länder. Debt and deficit figures from the Institute of International Finance (IIF), used for scale, are general-government concepts and are therefore larger. The two are not interchangeable, and this paper does not mix them in any single computation.

Japan requires separate treatment. The WDI interest series has no Japanese observation at all, and Japan’s debt stock, about 224 percent of GDP in Q4 2024 on IIF data, is far outside the range of the other four. The Japanese analysis uses the general-account budget, the part of the budget the Diet votes annually, and MOF bond-issuance plans, which distinguish new bonds from refunding bonds.

3 Data

3.1 Sources

Table 3 lists the sources. The fiscal core is the World Bank WDI, retrieved via the Elkassabgi-Data API in September 2026 (the World Bank API is the underlying source; ElkassabgiData is the retrieval interface). Four indicators are used: interest payments as a percent of revenue (GC.XPN.INTP.RV.ZS), revenue excluding grants as a percent of GDP (GC.REV.XGRT.GD.ZS), expense as a percent of GDP (GC.XPN.TOTL.GD.ZS), and central-government debt as a percent of GDP (GC.DOD.TOTL.GD.ZS). All are central-government concepts in current local currency units, converted to ratios.

Bond yields are annual averages of 10-year government bond series: the US Federal Reserve Board H.15 series, the German Bundesbank series, the Sveriges Riksbank series for Japan, and the IMF International Financial Statistics series for India, all retrieved through the ElkassabgiData library. No Chinese 10-year series was available in the library; the paper uses the People’s Bank of China policy rate path instead.

Debt scale comes from the IIF Global Debt Monitor for Q4 2024. US gross issuance comes from SIFMA’s 2024 fixed-income statistics. Japan fiscal data come from MOF budget documents: the FY2025 budget fact sheet, the FY2026 initial budget, the January 2025 MOF estimate reported by Reuters, and the Fiscal System Council’s high-yield scenario. Historical Japanese actuals are from the Board of Audit of Japan.

3.2 Coverage and sample period

Table 1 shows coverage by indicator and country. The WDI panel is unbalanced. The United States and Germany have near-complete series from the early 1970s through 2024. India’s interest and revenue series run 1974–2022 with a gap in 2019–2021. China’s interest series covers only 2005–2021, and China has no WDI expense series at all, so no balance or primary-balance computation is possible for China. Japan has no WDI interest series and almost no revenue series; it is analyzed from MOF sources only. Expenditure, needed for the balance, ends in 2023 for the US and 2022 for Germany and India, so balance-based metrics end one to two years before the interest series. The WDI central-government debt series is sparse outside the US and is not used in any computation; debt scale uses IIF throughout.

The analysis window shown in figures is 2000–2026. Computations use all available years. Data vintage: September 2026.

Table 1: Fiscal data coverage by country and indicator (WDI, central government)

Indicator	United States	China	Germany	India
Interest, % of revenue	1972–2024	2005–2021	1972–2024	1974–2022*
Revenue, % of GDP	1972–2024	2005–2024	1972–2024	1974–2022*
Expense, % of GDP	1972–2023	n/a	1972–2022	1974–2022*

*India has no WDI observations for 2019–2021. Japan is excluded: WDI carries no Japanese central-government interest series. Source: World Bank WDI via ElkassabgiData, retrieved September 2026.

3.3 Summary statistics

Table 2 reports the latest available observation for each country and indicator. The dispersion is the point: in the latest year, interest absorbs anywhere from 2.5 percent of revenue (Germany) to 34.0 percent (India).

Table 2: Latest available debt-service indicators

Country	Year	Interest % of revenue	Interest % of GDP	Govt debt % of GDP [†]
United States	2024	20.3	3.6	120
China	2021	3.4	0.5	92
Germany	2024	2.5	0.7	60
Japan	FY2025	9.2 [‡]	1.7	224
India	2022	34.0	3.1	83

[†]IIF Global Debt Monitor, Q4 2024, general-government concept; not comparable to the central-government ratios. [‡]Percent of general-account spending (MOF FY2025 initial budget), not of revenue. Source: WDI via ElkassabgiData; MOF; IIF.

4 Methodology

All computations are arithmetic on the series in Table 3; no statistical estimation is performed. For each country c and year t with interest share of revenue $i_{c,t}^{rev}$ and revenue share of GDP $r_{c,t}$:

$$i_{c,t}^{gdp} = \frac{i_{c,t}^{rev}}{100} \times r_{c,t} \quad (1)$$

which converts the interest burden to percent of GDP. The fiscal balance is proxied as revenue minus expense,

$$b_{c,t} = r_{c,t} - e_{c,t}, \quad (2)$$

where $e_{c,t}$ is expense as a percent of GDP. This is a net lending/borrowing proxy, not the WDI cash surplus/deficit series (GC.BAL.CASH.GD.ZS), which was not retrieved. The primary balance adds back interest:

$$p_{c,t} = b_{c,t} + i_{c,t}^{gdp}. \quad (3)$$

Net new borrowing is $n_{c,t} = -b_{c,t}$ when $b_{c,t} < 0$, and zero otherwise. The share of net new borrowing absorbed by interest is

$$s_{c,t} = \frac{i_{c,t}^{gdp}}{n_{c,t}}, \quad (4)$$

defined only in deficit years. Readings above 100 percent mean the government ran a primary surplus and still had to borrow to cover interest. The cumulative borrowing mix sums interest paid and primary deficits from 2015 through the latest year with complete data and reports interest as a share of the two combined.

Bond yields are annual averages of the underlying monthly or daily 10-year series. Japan's interest history combines Board of Audit actuals (FY1992, FY1999, FY2005), MOF initial budgets (FY2023, FY2025, FY2026), the MOF January 2025 projection for FY2028, and the Fiscal System Council high-yield scenario for FY2035. The FY2026 budget figure assumes the MOF's assumed interest rate rising from 2.6 to 3.0 percent. The yen figures are general-account interest payments in trillions of yen.

5 Results

5.1 The interest burden

Table 4 and Figure 1 give the headline result. The US federal interest burden doubled in four years, from 10.9 percent of revenue in 2020 to 20.3 percent in 2024, or 3.6 percent of GDP. The increase is entirely a rate effect working through refinancing: the debt stock was already large, and as low-coupon bonds matured they were replaced at higher yields. Figure 2 pairs the burden with the cause. The US 10-year annual average rose from 0.89 percent in 2020 to 4.21 percent in 2024;

Table 3: Data sources

Item	Source
Fiscal series	World Bank WDI, central government (interest, revenue, expense), via ElkassabgiData API, retrieved September 2026
10-year yields (US, Germany, Japan, India)	ElkassabgiData library: Fed Board, Bundesbank, Riksbank, IMF IFS; annual averages computed by author
China rates	PBoC policy rate path (no 10-year series in library)
Government debt, % of GDP	IIF Global Debt Monitor, Q4 2024
US gross issuance	SIFMA, 2024 fixed-income statistics
Japan budgets and projections	MOF FY2025 budget fact sheet; MOF FY2026 initial budget; MOF January 2025 estimate (via Reuters); Fiscal System Council high-yield scenario; Board of Audit of Japan (historical actuals)
US FY2024 deficit	Congressional Budget Office

Germany’s went from negative (-0.24 percent) to 2.65 percent; Japan’s from 0.00 to 0.89 percent and then 1.53 percent in 2025.

Table 4: Interest burden, latest year versus 2020

Country	Latest year	Interest % of revenue	Interest % of GDP	2020 interest % of revenue
United States	2024	20.3	3.6	10.9
China	2021	3.4	0.5	3.9
Germany	2024	2.5	0.7	1.0
India	2022	34.0	3.1	n/a*

*WDI has no Indian observation for 2020. Japan is in Section 5.4. Source: WDI via ElkassabgiData, author’s computations.

India is the extreme case. At 34.0 percent of revenue in 2022, more than a third of central-government revenue is pre-committed to interest before a single rupee of discretionary spending. The level has been above 20 percent for the entire 2000–2022 window shown in Figure 1, peaking above 40 percent in the early 2000s. Germany is the opposite: the burden fell steadily as rates declined, bottoming near 1.0 percent of revenue in 2020, and has only begun to turn up, reaching 2.5 percent in 2024. China’s burden is small and flat, 3.4 percent of revenue in 2021, and its monetary stance diverged: the PBoC eased while the Fed, the ECB, and eventually the Bank of Japan tightened.

5.2 How much new borrowing just pays interest

Table 5 reports, for the latest year with complete revenue and expense data, the decomposition of net new borrowing. In the United States in 2023, interest was 3.15 percent of GDP and net borrowing 6.94 percent of GDP, so 45 percent of every new dollar borrowed just paid the interest bill; the rest covered a primary deficit of 3.79 percent of GDP. In India in 2022, 73 percent of net

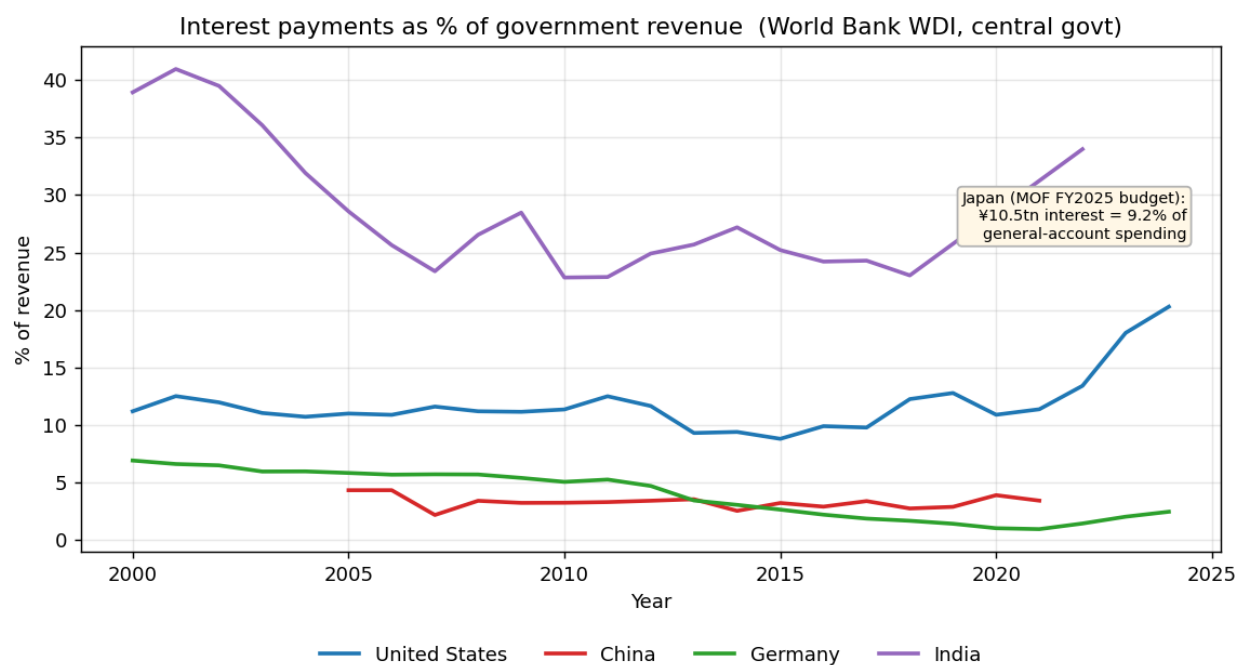


Figure 1: Interest payments as percent of central-government revenue, 2000–2024. India (purple) is persistently the highest, ending at 34.0 percent in 2022. The United States (blue) is flat near 10 percent until 2020, then rises steeply to 20.3 percent in 2024. Germany (green) falls to about 1 percent and ticks up to 2.5 percent. China (red) stays between 2 and 4 percent; its series ends in 2021. The Japan annotation is from the MOF FY2025 budget, not WDI. Source: World Bank WDI via ElkassabgiData; MOF.

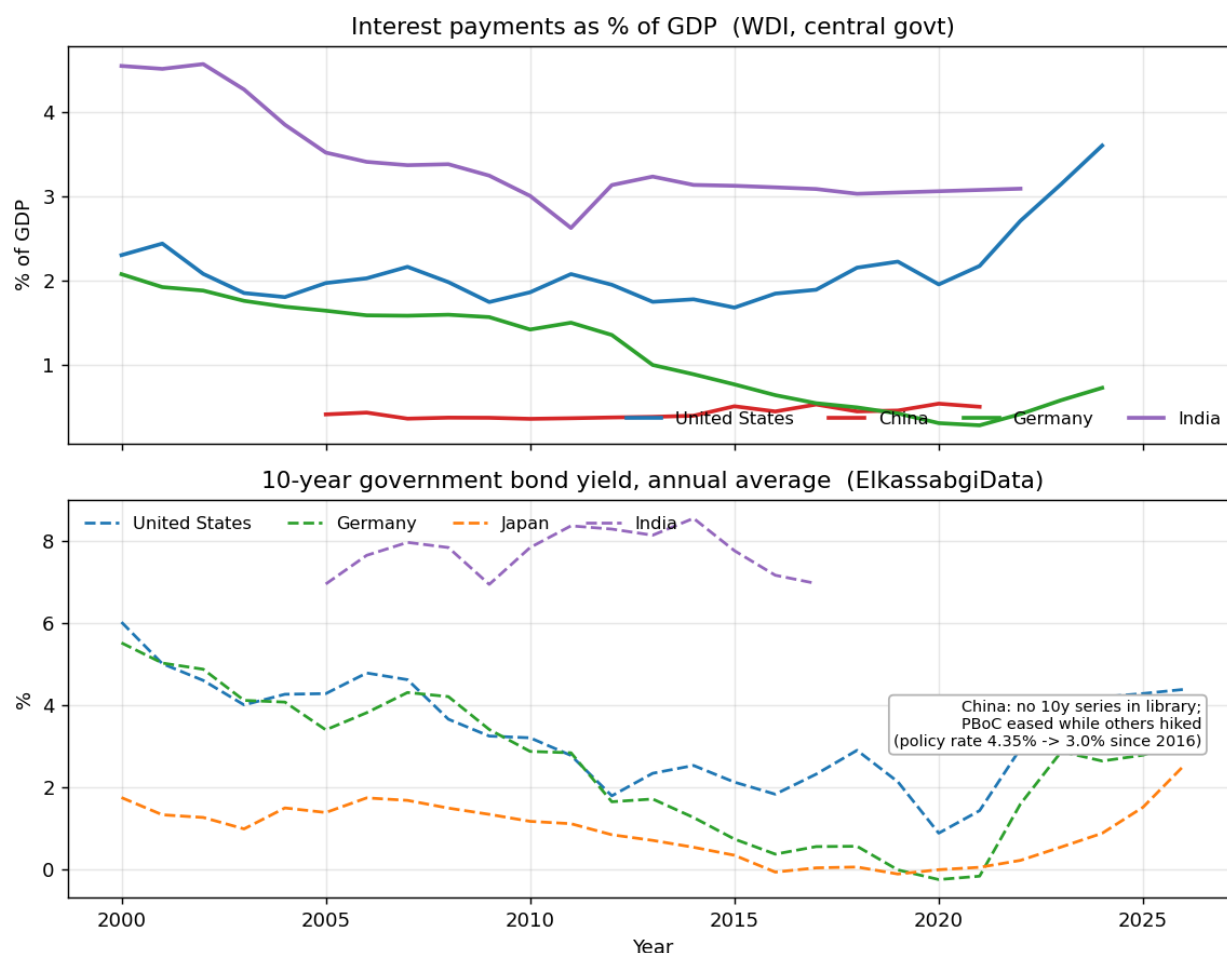


Figure 2: Top: interest payments as percent of GDP, central government. Bottom: 10-year government bond yields, annual averages. The US 10-year rose from 0.89 percent in 2020 to 4.21 percent in 2024; Germany from -0.24 to 2.65 percent; Japan from 0.00 to 0.89 percent. India's yield series ends in 2017 (7.17 percent in 2016). China has no 10-year series in the library; the PBoC cut its policy rate from 4.35 to 3.0 percent after 2016 while others hiked. Source: WDI via ElkassabgiData; ElkassabgiData yield library (Fed Board, Bundesbank, Riksbank, IMF IFS).

borrowing was interest, and the primary deficit was only 1.13 percent of GDP: nearly the whole deficit was the interest bill. Germany in 2022 was at 16 percent.

Table 5: Interest share of net new borrowing, latest year with full data

Country	Year	Interest % of GDP	Net borrowing % of GDP	Interest share of borrowing	Primary balance % of GDP
United States	2023	3.15	6.94	45%	-3.79
Germany	2022	0.42	2.61	16%	-2.19
India	2022	3.10	4.22	73%	-1.13

Deficit years only. China has no WDI expense series; Japan is in Section 5.4. Source: WDI via ElKassabgiData, author's computations.

Figure 3 plots the series from 2000. Three features stand out. First, readings above 100 percent are common for India and occurred for Germany in the 2000s: in those years the government ran a primary surplus and still borrowed, because interest exceeded the surplus. The debt was growing while the non-interest budget was in surplus. Second, the US series collapsed in 2020, not because interest fell but because the COVID deficit was enormous: interest was a small share of a very large borrowing number. By 2023 the share was back to 45 percent. Third, Germany's long decline from above 100 percent to 16 percent is the fiscal dividend of the low-rate era, now reversing at the margin.

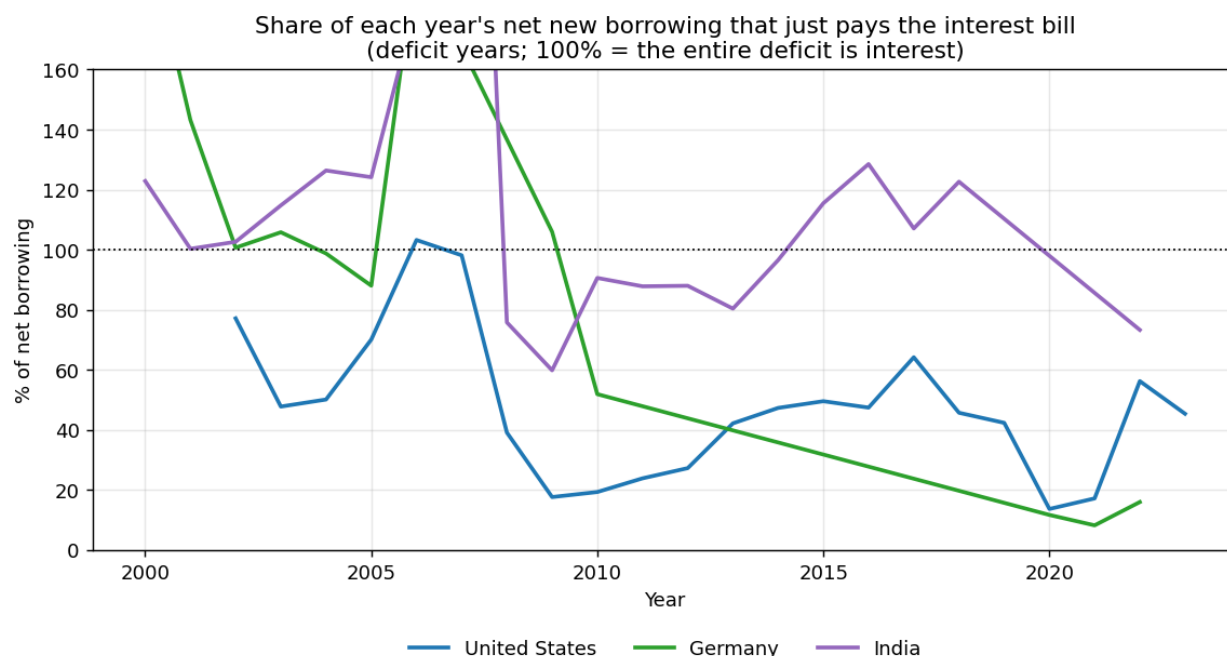


Figure 3: Share of each year's net new borrowing absorbed by interest payments, 2000–2023, deficit years only. The dotted line at 100 percent marks the point where the entire deficit is the interest bill; readings above it mean a primary surplus was insufficient to cover interest. India (purple) repeatedly exceeds 100 percent. The US (blue) share collapsed in 2020 on the COVID deficit, then recovered to 45 percent in 2023. Germany (green) declined from above 100 percent in the 2000s to 16 percent in 2022. Source: WDI via ElKassabgiData, author's computations.

5.3 The cumulative picture since 2015

Figure 4 (right panel) cumulates interest paid and primary deficits from 2015 through the latest complete year and reports interest as a share of total borrowing need. For the United States over 2015–2023, interest was 34 percent of all borrowing need (19.8 percent of GDP in cumulative interest against 39.2 percent of GDP in cumulative primary deficits). Germany over 2015–2022 is also 34 percent (3.9 against 7.7). India over 2015–2022 is 93 percent: cumulative interest of 15.5 percent of GDP against cumulative primary deficits of only 1.1 percent of GDP. India’s debt growth since 2015 is, to a first approximation, the accumulation of its interest bill.

The left panel of Figure 4 gives the debt scale for context, from IIF Q4 2024 data: Japan 224 percent of GDP, the United States 120 percent, China 92 percent, India 83 percent, Germany 60 percent. These are general-government figures and are not directly comparable to the central-government ratios elsewhere in the paper; they are shown to fix the stocks behind the flows.

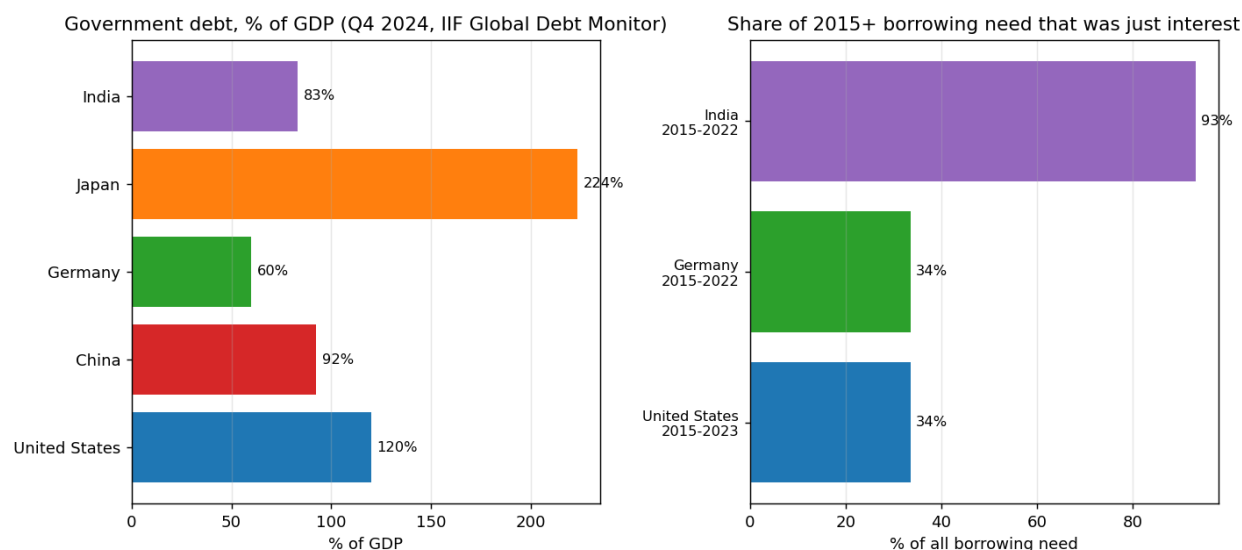


Figure 4: Left: government debt as percent of GDP, Q4 2024 (IIF, general-government concept). Right: interest as a share of total borrowing need, cumulated from 2015 through the latest year with complete data. India: 93 percent of 2015–2022 borrowing need was interest. The United States (2015–2023) and Germany (2015–2022): 34 percent each. Source: IIF Global Debt Monitor; WDI via ElkassabgiData, author’s computations.

5.4 Japan: the long view

Japan’s general-account interest payments trace the full arc of the low-rate era and its end (Figure 5). Actual payments were JPY 10.6 trillion in FY1992 and JPY 10.5 trillion in FY1999, then fell to JPY 7.0 trillion by FY2005 as rates declined, even as the debt stock grew. The FY2023 initial budget put interest at JPY 8.5 trillion; the FY2025 budget at JPY 10.5 trillion, which is 9.2 percent of general-account spending and about 1.7 percent of GDP. The FY2026 initial budget reaches JPY 13.0 trillion on an assumed interest rate rising from 2.6 to 3.0 percent. MOF’s January 2025 estimate projects JPY 16.1 trillion by FY2028. The Fiscal System Council’s high-yield scenario, in which rates normalize further, reaches JPY 45.2 trillion by FY2035, more than four times the FY2025

level.

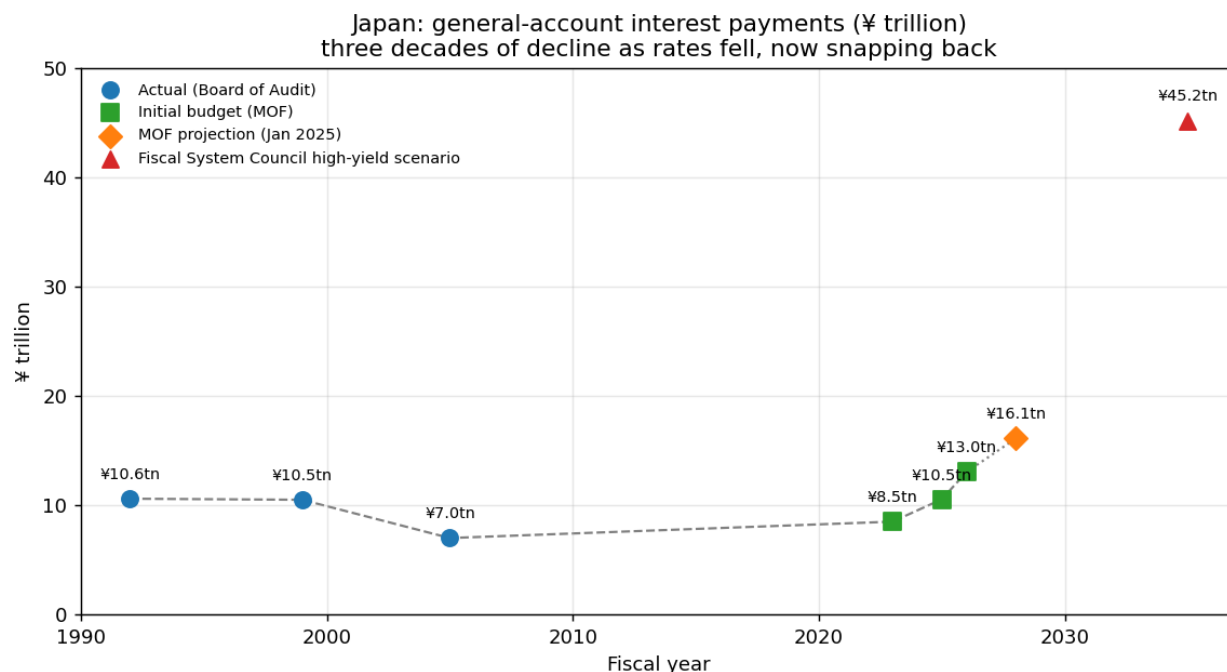


Figure 5: Japan general-account interest payments, trillions of yen. Actuals (Board of Audit): FY1992 JPY 10.6tn, FY1999 JPY 10.5tn, FY2005 JPY 7.0tn. Initial budgets (MOF): FY2023 JPY 8.5tn, FY2025 JPY 10.5tn, FY2026 JPY 13.0tn. MOF projection (January 2025): FY2028 JPY 16.1tn. Fiscal System Council high-yield scenario: FY2035 JPY 45.2tn. Three decades of decline as rates fell, now reversing. Source: Board of Audit of Japan; MOF; Reuters.

Two structural facts frame the Japanese numbers. First, roughly 83 percent of Japanese government bond issuance is refunding bonds, rolled principal, not new borrowing. The interest bill is the fiscal cost of the debt stock; the principal is perpetually refinanced. Second, the debt stock is about 224 percent of GDP (IIF, Q4 2024). At that stock, each percentage point of the effective interest rate is worth more than 2 percent of GDP in annual debt service, which is why the MOF projections accelerate so fast once the assumed rate moves.

5.5 Refinancing: gross versus net issuance

The principal side of debt service is refinancing, everywhere in the sample. The cleanest quantification is the United States, where SIFMA reports 2024 gross federal issuance of USD 29.3 trillion against a FY2024 federal deficit of about USD 1.8 trillion (CBO), the net new borrowing. Roughly 93 percent of gross issuance therefore refinanced maturing principal; about 7 percent was new debt. The same arithmetic holds qualitatively for the other four: no large sovereign repays principal from revenue at any scale. Debt service, as a fiscal burden, is the interest bill. The principal is a rollover operation, and its risk is refinancing risk, the risk that markets will not roll the stock at tolerable rates, not repayment risk.

6 Limitations and robustness

The WDI fiscal series cover the central government only. State, prefectural, and municipal debt and interest are excluded, so the burdens measured here understate general-government burdens, materially for Japan and Germany, where subnational borrowing is significant.

Coverage gaps are material. China has no WDI expense series, so no balance, primary balance, or interest-share-of-borrowing computation exists for China; the Chinese results are interest ratios only, ending in 2021. Japan has no WDI interest series at all and is analyzed from MOF budget documents, which are general-account concepts, not central-government WDI concepts; the two are not strictly comparable. India's WDI series skip 2019–2021, so no 2020 comparison is available for India. India's 10-year yield series ends in 2017, so the yield panel cannot show India's recent rate path.

The balance is proxied as revenue minus expense, not the WDI cash surplus/deficit series, which was not retrieved. For central governments the two track closely, but they can differ in years with large net acquisition of financial assets.

The Japanese FY2026 budget figure and the FY2028 projection come via MOF documents and Reuters reporting of MOF estimates; they are official projections, not outturns, and the FY2028 number predates the FY2026 budget. The Fiscal System Council FY2035 figure is a high-yield scenario, an illustration of rate sensitivity, not a forecast. The 83 percent refunding share is from MOF issuance plans and moves year to year.

All WDI data are as retrieved in September 2026 and are subject to revision by the World Bank and by national sources. The IIF debt figures are Q4 2024; the SIFMA issuance figure is full-year 2024.

7 Conclusion

The five largest economies fund debt service the same way: they refinance principal and pay interest from revenue supplemented by new borrowing. What differs is how much of the revenue interest consumes. The 2021–2024 hiking cycle doubled the US burden to 20.3 percent of revenue; India was already at 34.0 percent and has been above 20 percent for two decades; Germany, at 2.5 percent, shows what the low-rate era bought and what its end is starting to take back; China is small and easing against the global tide; Japan's bill, after thirty years of decline, is projected to rise by half again by FY2028.

The binding constraint in every case is the interest share of revenue and the interest share of new borrowing, not the debt stock as a number. A debt stock of 224 percent of GDP at near-zero rates cost Japan less, relative to its budget, than India's far smaller stock costs India today. When rates move, the arithmetic in this paper is the mechanism.

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A Data appendix

A.1 Series definitions

Table 6: WDI series used

Label	WDI code	Definition
int_rev	GC.XPN.INTP.RV.ZS	Interest payments, % of revenue
rev_gdp	GC.REV.XGRT.GD.ZS	Revenue excluding grants, % of GDP
exp_gdp	GC.XPN.TOTL.GD.ZS	Expense, % of GDP
debt_gdp	GC.DOD.TOTL.GD.ZS	Central government debt, total, % of GDP

Retrieved via ElkassabgiData (worldbank series), September 2026. debt_gdp is documented for completeness; the paper uses IIF debt figures for scale and does not compute from debt_gdp.

A.2 Derived series

For each country-year with complete inputs: interest as percent of GDP = $\text{int_rev}/100 \times \text{rev_gdp}$; balance = $\text{rev_gdp} - \text{exp_gdp}$; primary balance = balance + interest % of GDP; net borrowing = $-\text{balance}$ in deficit years; interest share of borrowing = interest % of GDP / net borrowing. Cumulative borrowing mix: sums from 2015 through the latest year with complete revenue and expense data.

A.3 Source files

The local source files are the `econ_*.csv` pulls (one per country-indicator, ElkassabgiData WDI downloads, September 2026), the `wdi_fiscal_long.csv` panel, and the yield CSVs in the bond-market-trends data directory. Japan figures use the MOF, Board of Audit, and Reuters documents listed in References. Replication code: `analyze2.py` (figures 1–4 and all WDI-derived tables), `japan_fig.py` (figure 5), `fetch_wdi.py` (WDI retrieval).